

Mass Segregation in the CMZoom Survey

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ABSTRACT

We employ a Minimum Spanning Tree (MST) approach to characterize the spatial distribution and mass segregation of compact millimeter continuum sources within the Central Molecular Zone (CMZ) of the Milky Way. We use a modified form of the complete version of the 1.3 mm dust continuum catalog from the CMZoom survey, which identifies 685 compact sources with typical effective radii of ~ 0.1 pc. For 22 of 35 CMZ clouds, we calculate the thermal and turbulent Jeans lengths and masses, and determine that compact source separations, as well as compact source masses, are more consistent with thermal fragmentation at ~ 0.1 pc size scales. We construct the mass segregation ratios for compact sources in 17 CMZ clouds and determine that 5 of the analyzed clouds display some form of mass segregation ($\Lambda_{MSR} > 1.5$), while the remaining clouds show either inverse mass segregation ($\Lambda_{MSR} < 0.75$), or no evidence of true mass segregation ($0.75 < \Lambda_{MSR} < 1.5$). Finally, we find that although some actively star-forming clouds do exhibit mass segregation, other similarly active clouds do not, indicating an unclear correlation with evolutionary stage for star forming clouds in the CMZ, given the current available data.

1. INTRODUCTION

The Central Molecular Zone (CMZ) consists of the inner few hundred pc of the Galactic Center (Henshaw et al. 2022; Battersby et al. 2025a), and contains roughly 5% of the molecular gas in the Milky Way (Dahmen et al. 1998), most of which is concentrated in the numerous dense molecular clouds such as G0.699-0.028 (Sgr B2), G359.484-0.132 (Sgr C), and the Dust Ridge (Mills et al. 2018). It is an extreme environment, with a high gas number density of $n_{\text{H}_2} > 10^4 \text{ cm}^{-3}$ (e.g., Paglione et al. 1998; Jones et al. 2013), extreme temperatures > 60 K in comparison to the Galactic disk (Ginsburg et al. 2016), and magnetic field strengths of ~ 1 mG for some clouds (Mangilli et al. 2019; Federrath et al. 2016). Lu et al. (2024) find magnetic field strengths in several molecular clouds in the CMZ to range between 0.1 mG to 1.73 mG.

Despite the abundance of dense gas, the CMZ is widely considered to be under-producing stars overall (Kauffmann et al. 2017; Immer et al. 2012; Guesten & Downes 1983; Beuther et al. 2012; Barnes et al. 2017). Using the CMZoom survey, Hatchfield et al. (2020) put a lower and upper limit on the potential for star-formation

in the CMZ of ≈ 0.04 to $0.47 M_{\odot} \text{ yr}^{-1}$ (with the exclusion of G0.699-0.028; ≈ 0.08 to $2.20 M_{\odot} \text{ yr}^{-1}$ inclusive of G0.699-0.028). Hatchfield et al. (2024) argue that the incipient star formation rates (SFRs) in the CMZ might in fact be higher than previously estimated, due to the fact that use of the cold dust continuum alone cannot fully differentiate between active star-formation and quiescent dust, and calculate limits similar to those found by Hatchfield et al. (2020).

We study mass segregation within star-forming clouds in the CMZ as selected by the CMZoom Survey (Battersby et al. 2020; Hatchfield et al. 2020). Prior studies have investigated the mass segregation of stellar clusters to impose constraints on their formation and evolution (Pouteau et al. 2023; Pavlík et al. 2019; Dib & Henning 2019). These studies have used a variety of methods such as the mass segregation ratio (MSR), which uses minimum spanning trees (MSTs) to characterize mass distribution in a cluster (Allison et al. 2009).

More recently, these analyses have been performed on clouds that are forming pre- and protostellar cores (Kinman et al. 2024; Morii et al. 2023). Nony et al. (2021) in particular, use Allison et al. (2009)’s method to calculate the mass segregation for clouds and concludes that initial mass segregation of gas in dense prestellar cores

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may contribute to the eventual mass segregation of stars in clusters. [Sanhueza et al. \(2019\)](#) find no strong evidence of early (primordial) mass segregation of prestellar cores in the infrared dark clouds surveyed, but do conclude that future core accretion may result in dynamic core mass segregation. [Dib & Henning \(2019\)](#) examine (prestellar and dense) cores in four star-forming clouds and conclude that three of those clouds exhibit primordial mass segregation, where the cores initially form within the densest parts of the clouds.

The current state of knowledge on the mass-distribution of compact sources in the CMZ is limited to a few well-researched clouds such as the G0.699-0.028 cloud, G359.484-0.132 cloud, and G0.253+0.016 (the Brick) ([Kinman et al. 2024](#)). When examining millimeter CMZ sources on the scale of ≈ 1000 au, [Kinman et al. \(2024\)](#) found that G359.484-0.132 shows signs of mass segregation and star-formation, while G0.253+0.016 shows signs of neither, and make note that the most massive prestellar cores observed are expected to be located in the densest parts of the cloud, in order to facilitate future accretion.

In order to gain a more complete understanding of compact source mass segregation in the CMZ, we need to leverage millimeter-wavelength observations of the dust continuum at sub-parsec scales that are sufficient to resolve the individual star-forming clouds throughout the entire CMZ. The *CMZoom* survey is the first large-scale survey of the CMZ at arcsecond / sub-parsec resolution ([Battersby et al. 2020](#)). This makes it a good candidate for studying mass segregation throughout the densest star-forming clouds of the CMZ in a uniform manner. Future studies, such as the ALMA CMZ Exploration Survey, ACES, will help further these analyses ([Longmore et al. 2026](#)).

In this paper, we investigate the distribution and mass segregation of compact sources for a complete sample of star-forming clouds located in the Central Molecular Zone. In Section 2 we describe the *CMZoom* Survey SMA observations and the *CMZoom* compact source catalog. We describe our method for characterizing fragmentation throughout the CMZ in Section 3 and our MSR analysis using MSTs in Section 4. We report the results of our analyses in Section 5 and discuss the broader implications of our results in Section 6.

2. SUMMARY OF DATA

2.1. SMA Observations

The *CMZoom* Survey is a large Submillimeter Array (SMA) program that observed all of the high-column-density gas ($N(\text{H}_2) > 10^{23} \text{ cm}^{-2}$) contained within the CMZ ([Battersby et al. 2020](#); [Hatchfield et al. 2020](#)). The

1.3 mm continuum emission data have an angular resolution of $\sim 3''$, corresponding to a typical spatial resolution of ~ 0.1 pc at an assumed distance of 8.178 kpc ([GRAVITY Collaboration et al. 2019](#)). A total of 35 clouds were observed covering an area of ~ 240 square arcminutes ($\sim 1360 \text{ pc}^2$). All but 4 of the 35 clouds (G1.670-0.130, G0.393-0.034, G0.212-0.001, and G359.137+0.031) are confirmed to be located in the CMZ by [Hatchfield et al. \(2024\)](#), and [Battersby et al. \(2020\)](#) find that G0.316-0.201 and G359.615-0.243 are not in the CMZ. Additionally, the *CMZoom* program observed spectral line emission from a variety of molecular species, including CO, H_2CO , and CH_3OH ([Callanan et al. 2023](#)).

2.2. CMZoom Compact Continuum Source Catalog

The analysis we perform in this paper makes use of a version of the complete *CMZoom* Survey compact dust continuum source catalog, with an original total of 816 sources ([Hatchfield et al. 2020](#)). Since the primary beam corrected SMA images have enhanced noise at the edges, we remove catalog sources within 0.5 pc of the map boundary, to reduce the impact of possible false detections on our mass segregation analysis. This reduced version of the complete *CMZoom* catalog has a total of 685 sources. This catalog was generated using a pruned dendrogram-based technique, *astrodendro*¹, which is a Python package that takes astronomical data and decomposes it into a dendrogram, or a hierarchical representation of the data. The dendrogram is a tree-like diagram where the densest structures that contain no additional substructures are categorized as “leaves”, whereas structures containing additional substructure are known as “branches” and “trunks”. The only structures included in the *CMZoom* catalog are the densest “leaf” structures (or “compact sources”), with typical effective radii between 0.04 and 0.4 pc at an assumed distance of 8.178 kpc.

The dendrogram source extraction performed by [Hatchfield et al. \(2020\)](#) required structures to have a minimum pixel value larger than $3\sigma_{\text{global}}$ and a minimum leaf boundary significance of $1\sigma_{\text{global}}$, where σ_{global} is the minimum root-mean-squared (RMS) noise estimate for the entire surveyed cloud, 3 MJy sr^{-1} . Each structure contains a minimum number of pixels equivalent to half the typical beam size. Additionally, the cataloged leaf structures were required to meet local noise thresholds. The local RMS noise estimates were calcu-

¹ Refer to <http://www.dendrograms.org/> for additional information on the *astrodendro* package, and to [Hatchfield et al. \(2020\)](#) section 4.3, Appendices A and B for further information on the chosen parameters.

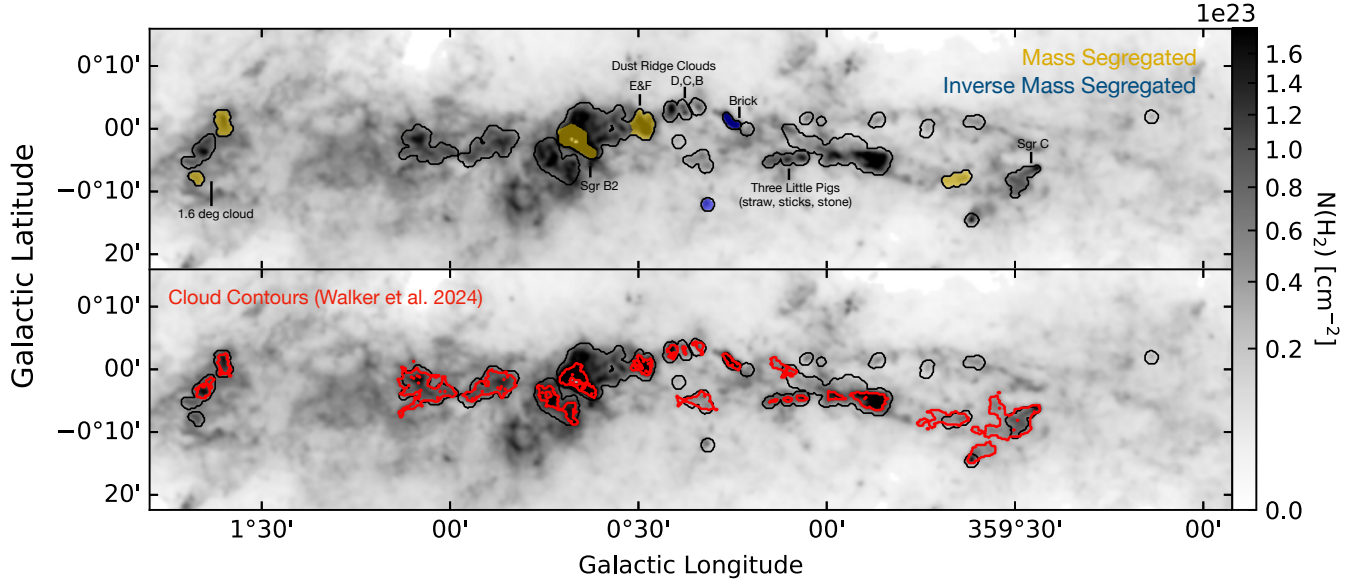


Figure 1. An overview of the clouds covered in the CMZoom survey. The Herschel Hi-GAL column density map of the CMZ (Battersby et al. 2024) is shown in gray scale, and black contours represent the area covered by the CMZoom Survey. The top panel describes each cloud by its level of mass segregation as determined from its Mass Segregation Ratio (MSR) in Section 4.1. Regions that are shaded yellow are considered mass segregated, whereas the blue shaded regions are considered inverse mass segregated. Unshaded regions represent the clouds that do not have enough source detections to make a meaningful MSR calculation or can represent regions that do not meet the MSR thresholds for (inverse) mass segregation. A number of well-known CMZ clouds are labeled using their colloquial names. The bottom panel shows the CMZ clouds defined in (Walker et al. 2024, red contours) that are associated with our CMZoom regions, which we use to measure thermal and turbulent Jeans properties in Section 3.

lated using SMA residual maps, as detailed in Battersby et al. (2020). The complete version of the “pruned” CMZoom catalog only includes sources with a peak flux of at least $4\sigma_{local}$, while the robust catalog requires sources to have a peak flux of at least $6\sigma_{local}$. Both the complete and robust catalogs exclude leaves with a mean flux $< 2\sigma_{local}$. While the complete catalog is sensitive to lower mass sources and most of the sources unique to the complete catalog are in the low-mass range and contain roughly 13% of the mass of the robust catalog, there is a higher frequency of false positive detections, although Hatchfield et al. (2020) finds that a significant number of false positives are unlikely in the lower-mass range. Conversely, the robust catalog prioritizes the reduction of false positive detections, but increases the chance of removing real, lower-mass sources. In this paper, we use the reduced form of the complete CMZoom catalog to ensure our analysis neither excludes low mass detections nor contains too many false positives. However, we also consider the effects of including only the most robust detections in our analysis, as discussed in Section 5.1.1. For a more detailed description of the applied dendrogram source extraction of the *CMZoom*

continuum data, we refer the reader to Hatchfield et al. (2020).

2.3. CMZ Cloud Catalog

In Section 3, we use the dust temperatures, gas densities, and gas velocity dispersions for molecular clouds in the CMZ to characterize their fragmentation properties. Our analysis makes use of the CMZ cloud catalog presented in Walker et al. (2024) and Battersby et al. (2025b).

This catalog was generated from the Herschel dust column density map from Battersby et al. (2025a) using the Python package *astrodendro* as described in Battersby et al. (2025b), with specific selection criteria applied in Walker et al. (2024). Further information about the *astrodendro* input parameters and the precise implementation are provided in Walker et al. (2024); Battersby et al. (2025b). The molecular clouds reported in this catalog have effective radii ranging from 1.4 pc – 8.7 pc and masses ranging from $1 \times 10^4 - 1 \times 10^6 M_{\odot}$. The dust temperature map presented in Battersby et al. (2025b) was used to estimate the median and peak temperature for each cloud. Walker et al. (2024) derived cloud velocity dispersions by performing multiple com-

ponent Gaussian fits on HNC0 ($4_{0,4} - 3_{0,3}$) and H_2CO ($3_{0,3} - 2_{0,2}$) spectra². This is performed at angular resolutions ranging from $3''$ to $39''$ and spatial resolutions ≥ 4 pc, which are similar to or coarser than the *CMZoom* survey resolution. Although both species are reliable tracers of dense gas in the CMZ, we use the velocity dispersions derived from the H_2CO spectral line fits for our analysis. We find that H_2CO has fewer components per cloud than HNC0 in Walker et al. (2024), which simplifies the analysis. Nonetheless, the derived velocity dispersions for both H_2CO and HNC0 are similar enough in magnitude that the use of either tracer does not change the conclusions of our results.

2.4. Overview of Clouds and Tables

The *CMZoom* Survey looks at multiple clouds which are areas of intense research in the CMZ, including G0.699-0.028, G359.484-0.132, G0.253+0.016, and the Dust Ridge regions (G0.489+0.010, G0.412+0.052, G0.380+0.050, and G0.340+0.055). We are unable to measure the mass segregation ratios for certain clouds (such as G359.484-0.132) which have too few sources in our modified version of the complete compact catalog.

In this work, we use the non-background-subtracted masses for each source. Although this may be an overestimate of the true mass, we find that the choice in using the non-background-subtracted masses over the background-subtracted masses has a minimal impact on the classification of our clouds. The clouds that show the most significant change in MSR when using the background-subtracted sources instead of the non-background-subtracted sources can be seen in Appendix B, and are: G0.380+0.050 (Dust Ridge cloud C), G0.326-0.085, G0.316-0.201, and G359.889-0.093 (20 km/s cloud).

Table 1 contains the calculated values for both the mass segregation analysis and the Jeans lengths and masses. Table 2 compares whether specific clouds display both mass segregation and detected star formation tracers (from Hatchfield et al. (2024) and Battersby et al. (2025b), the latter of which uses the former’s estimates).

3. THERMAL AND TURBULENT JEANS LENGTH AND MASS

3.1. Minimum Spanning Tree Method

In order to investigate the type of fragmentation in the clouds included in this survey, we compare the mean edge length ($\bar{\ell}$) of the particular cloud’s minimum span-

ning tree (MST) to both the cloud’s thermal Jeans length, λ_{th} and its turbulent Jeans length, λ_{turb} .

An MST is the shortest path between all points in a group, where the individual paths between each point are known as “edge lengths” (Prim 1957; Barrow et al. 1985). The MST uniquely allows for a view of the spatial separation between the compact sources without relying on such other methods as mean angular separation.

In order to generate the MST for a cloud with more than two defined compact sources, we first build a sparse matrix composed of the angular separation (in degrees) between each combination of compact sources within a *CMZoom* cloud. When this sparse matrix has been built for the cloud, we apply the `scipy minimum_spanning_tree` package, which optimizes the shortest overall path using the distances in the matrix (see Figure 2).

The presence of distant outlier detections may affect the construction of the MST, such as its shape and total length, which ultimately affects our analysis of these regions. We discuss how these outlier detections may affect our subsequent results in Section 6.2.

Fig. Set 1. *CMZoom* Overview Figures

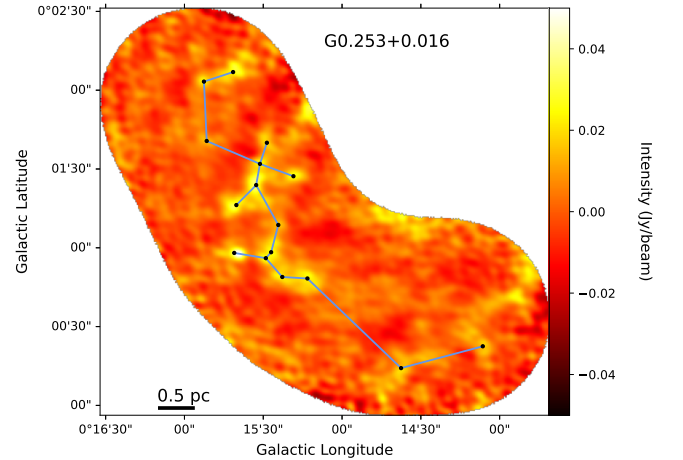


Figure 2. Applying the MST algorithm as discussed in Section 3.1 to the case of the G0.253+0.016 cloud (shown), a simple elongated path. The black points represent the locations of the compact sources based on their galactic coordinates. The blue lines connecting the black points are the edge lengths of the MST. The 1mm dust continuum SMA observation for the cloud is underlaid. The complete figure set (35 images) is available in the online journal.

3.2. Thermal Jeans Length and Mass

We follow Sanhueza et al. (2019)’s correction factor (division by $\frac{2}{\pi}$) to calculate the deprojected mean edge

² See Section 2 in Walker et al. (2024) for information on the specific data.

length ($\bar{\ell}_{\text{corr}}$), which we compare with λ_{th} and λ_{turb} . We report $\bar{\ell}_{\text{corr}}$ for each cloud in Column 8 of Table 1.

In this work, we follow the analysis of Zhang et al. (2021) with the equation for λ_{th} as follows, derived in Mac Low & Klessen (2004).

$$\lambda_{th} = \sigma_{th} \sqrt{\frac{\pi}{G\rho}} \quad (1)$$

where

$$\sigma_{th} = \sqrt{\frac{k_B T}{\mu m_H}} \quad (2)$$

We use a value of 2.37 for μ (the mean molecular weight per free particle; Kauffmann et al. 2008), Herschel Hi-GAL median dust temperatures (ranging between 20 and 25 K) for each cloud, and Herschel Hi-GAL densities as ρ .

We have also considered the thermal Jeans mass, M_{th} (Mac Low & Klessen 2004) as follows:

$$M_{th} = \frac{4\pi\rho}{3} \left(\frac{\lambda_{th}}{2} \right)^3 = \frac{\pi^{5/2}}{6} \frac{\sigma_{th}^3}{\sqrt{G^3\rho}} \quad (3)$$

In our calculations of the Jeans lengths and masses, we consider the main sources of error to be the dust temperature (a statistical error of roughly 10%, please see Section 6.2 for more about the systematic uncertainty in the core temperatures), and the densities from the Herschel Hi-GAL table (an uncertainty of a factor of 2) (Battersby et al. 2025a). These uncertainties have been included in the calculations of the Jeans lengths and masses, and are shown in Table 1.

3.3. Turbulent Jeans Length and Mass

The turbulent Jeans length, λ_{turb} , and turbulent Jeans mass, M_{turb} , are calculated in the same way as λ_{th} and M_{th} , but with non-thermal linewidths rather than sound speed (eg, in Wang et al. 2014). We calculate λ_{turb} and M_{turb} for every cloud, using the Gaussian σ widths converted from non-thermal FWHM linewidths for H₂CO from Walker et al. (2024), except for G0.316-0.201, which has no detected H₂CO lines.

The H₂CO linewidths from Walker et al. (2024) are derived at scales at or larger than that of the *CMZoom* survey. The derivation at this scale potentially overestimates the velocity dispersions of the gas enclosing the star-forming cores, given that velocity dispersions have been shown to decrease at smaller physical scales (Kauffmann et al. 2017). We therefore consider a “clump-scale”, which is defined as the smallest ellipse that encloses all sources within a region. The ellipses are applied to the column density and temperature maps produced by Battersby et al. (2025b); the column densities within the ellipse are averaged according to the

method given in Battersby et al. (2025b) to calculate the masses. This ellipse is approximated to be a prolate spheroid in order for the volume density to be calculated from the mass, and the median temperature is taken for the calculation of the sound speed at that size scale ($\approx 3 - 4$ pc). All further calculations pertaining to the thermal and turbulent Jeans lengths and masses still use Equations 2 and 3 from Section 3.2. The CMZ is understood to have steep linewidth-size relations (see: Kauffmann et al. 2017; Shetty et al. 2012), and the power-law $\sigma \propto R^{0.663}$ (Shetty et al. 2012; Kauffmann et al. 2017) is used to derive the non-thermal linewidths. See Figure 5 for comparison between clump-scale and cloud-scale properties.

4. MASS SEGREGATION RATIO

We characterize the mass segregation of compact continuum sources identified in the *CMZoom* catalog using the Mass Segregation Ratio (MSR), developed by Allison et al. (2009).

We use this MST-based method to define mass segregation instead of density profiles, mass functions or local density ratios because we choose to focus on source-to-source comparison across the regions. Further, this method does not rely on determining the “center” of the cloud, which, for methods such as the mass function, is already assumed to be the location of the most massive stars (Allison et al. 2009). This difference in definition complicates comparison with studies using different methods. For specifics on the differences between the methods, please see Allison et al. (2009) and Parker & Goodwin (2015).

We are unable to calculate the mass segregation ratios for every *CMZoom* survey cloud due to low number statistics (see Appendix A for details). A higher resolution dataset would be needed in order to study these excluded clouds; one of the excluded clouds is G359.484-0.132, which contains only 7 sources in our modified version of the complete catalog.

4.1. Mass Segregation Ratio: Arithmetic

We use the MSR because it is a quantification of the distribution of mass throughout a star-forming cloud that helps to illustrate their morphology. This calculation is complicated by the specific definitions used in this work for both the masses of the compact sources and the derivation of the mass; see Section 6.2.

³ A proportionality constant is not chosen here due to the lack of an appropriate fit for the scale-linewidth relation of the given regions.

The arithmetic Mass Segregation Ratio (Λ_{MSR}), with its standard deviation,

$$\Lambda = \frac{\langle \lambda_{rand} \rangle}{\lambda_{mass}} \pm \frac{\sigma_{rand}}{\lambda_{mass}} \quad (4)$$

is a ratio between the length of the MST for N number of the most massive compact sources in a cloud (λ_{mass}) and the (arithmetic) average MST length of K sets of N random compact sources ($\langle \lambda_{rand} \rangle$) Allison et al. (2009), where σ_{rand} is the standard deviation of $\langle \lambda_{rand} \rangle$. The MSR assumes nothing about the specific spatial geometry of a cloud (Parker 2018; Alcock & Parker 2019; Dib & Henning 2019). By our definition, an MSR > 1.5 indicates mass segregation, where the more massive compact sources are spatially close together, while an MSR < 0.75 indicates inverse mass segregation, with the more massive compact sources spatially farther apart. An MSR between the bounds shows a uniform mass distribution, with the mass spread evenly throughout the cloud (see Appendix A for details).

For our sample, we take K = 500 to be sufficient for our calculation of the mass segregation ratio (see Appendix A for an explanation for this choice in K). Some clouds in our sample do not meet the required number of compact sources for us to calculate the MSR with K=500 (see Appendix A) and are not considered in our MSR analysis. For each CMZoom cloud with ≥ 16 compact sources, we calculate N_{MST} for N = 3 to the last compact source in the cloud, N_{max} , and produce an MSR curve. We plot the MSRs for the CMZoom clouds, ranging from N = 3 to N = 20, with K kept constant at 500 in Figure Set 1. Further analysis of these figures is found in Section 5.1.

According to Allison et al. (2009), the averaging of λ_{rand} results in a roughly Gaussian dispersion, which is then quantified by the standard deviation as defined above (Equation 4). We have also followed this method of standard deviation in this work.

The MSR is very sensitive to different assumptions, such as dust temperature, in the compact source-mass estimate. For the catalogs used in this work, and as outlined in Hatchfield et al. (2020), the observed Herschel dust temperatures for the clouds range from 20K to 30K.

Since the MSR is dependent on the construction of the MST, it is sensitive to the inclusion of outlier detections, however, as discussed in Appendix A, our results are minimally impacted. A more complete discussion can be found in Section 6.2.

4.2. Mass Segregation Ratio: Geometric

We use the Geometric Mass Segregation Ratio (Γ_{MSR}) as developed by Olczak et al. (2011) and used in Dib & Henning (2019) and Dib et al. (2018):

$$\Gamma = \frac{\langle \gamma_{rand} \rangle}{\gamma_{mass}} \quad (5)$$

where γ is the geometric mean of the minimum spanning tree lengths, e_i :

$$\gamma = (\prod_{i=1}^n e_i)^{\frac{1}{n}} \quad (6)$$

and the standard deviation is geometric, as follows:

$$\Delta \gamma_{rand} = \exp \left(\sqrt{\frac{\sum_{i=1}^{N_{cs}-1} (\ln e_i - \ln \gamma_{rand})^2}{N_{core} - 1}} \right) \quad (7)$$

The K which we use for Γ_{MSR} is the same as for Λ_{MSR} , 500.

5. RESULTS

5.1. Arithmetic Mass Segregation in the Clouds

Typically, a region is considered mass segregated if it has an MSR > 1, and inverse mass segregated if it has an MSR < 1. For this work, we choose to enforce stricter criteria to limit the probability that a region is classified as (inverse) mass segregated by chance, where regions with MSR < 0.75 are inverse mass segregated, and regions with MSR > 1.5 are mass segregated. See Appendix A for details of our definition of mass segregation, as compared to the established threshold of MSR = 1.

In the MSR plots (as seen in Figure 3), the magnitude of mass segregation is shown on the y-axis and the x-axis shows each consecutive N_{MSR} used in the calculations for the MSR, beginning with $N_{MSR} = 3$.

We identify 1 cloud G0.316-0.201 that exhibits inverse mass segregation, or an MSR < 0.75, as shown in the bottom panel of Figure 3.

We consider clouds that exhibit an Λ_3 between 1.5 and 2.5 (without considering standard deviation) to be mildly mass-segregated (see Appendix A and the top-right and middle panels of Figure 3). We identify 3 clouds in this category, G0.699-0.028, G1.670-0.130, and G359.889-0.093. The G359.889-0.093 and G1.670-0.130 clouds show an increase in the degree of mass-segregation with the increase of N, starting from MSR ≈ 1.1 at N = 3 to MSR ≈ 1.6 at N = 5, with variation in the curves. The clouds do converge to MSR = 1, as expected. This increase in MSR and subsequent variation does not neatly fit into the definition of mass-segregation, and indicates a complex substructure

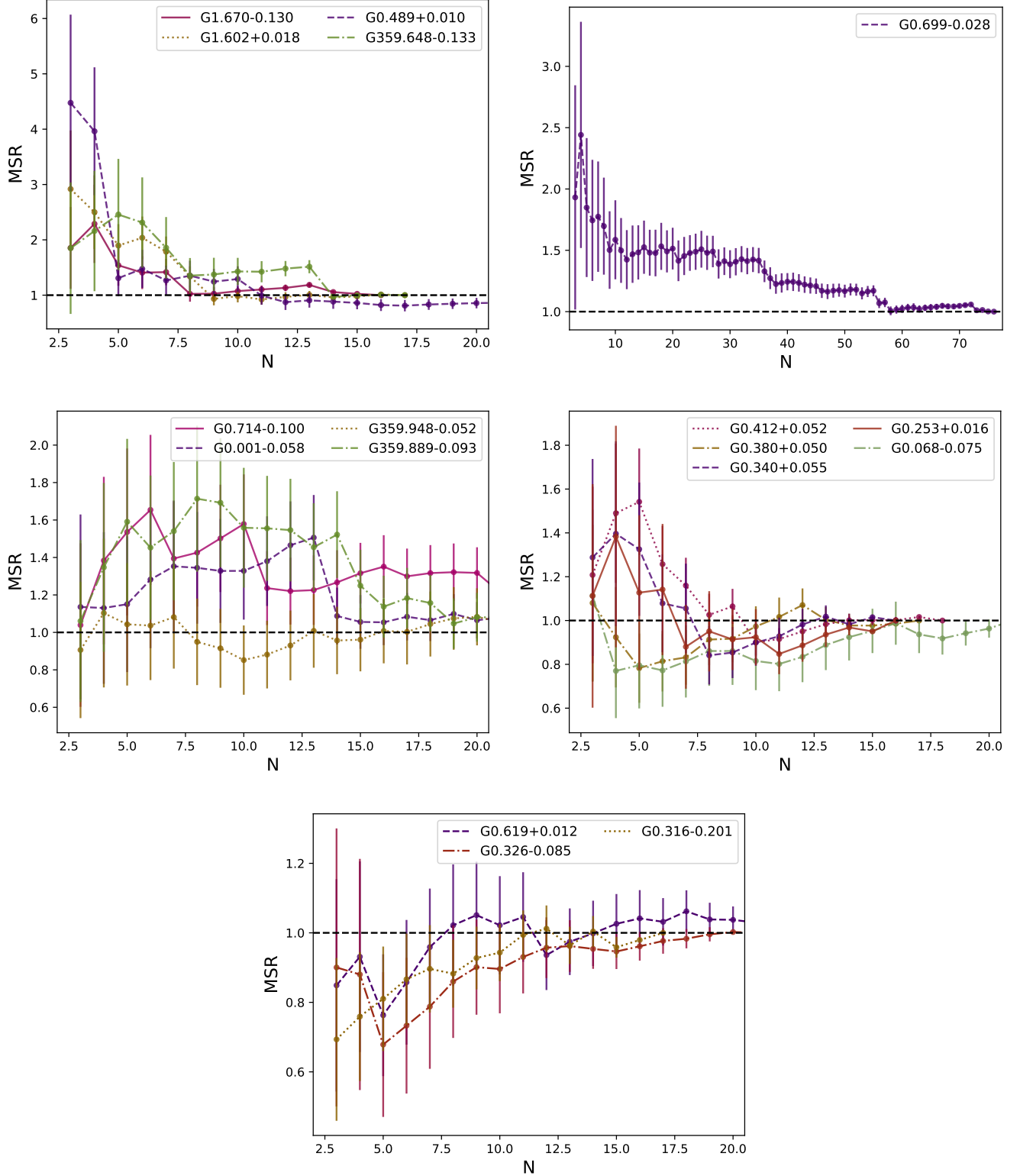


Figure 3. This figure displays the MSR curves for the 17 clouds included in this survey that have at least 16 compact sources detected. (*Top*) The MSR plots for the mass segregated clouds ($\Lambda_3 > 1.5$), with the full plot for G0.699-0.028. N is limited to $N = 20$ otherwise due to the tendency of the MSR to “level” at $\text{MSR} = 1$ for $\frac{N_{\text{total}}}{N_{\text{MSR}}} \rightarrow 1$. The dotted black line indicates $\text{MSR} = 1$. (*Middle*) The MSR plots for the clouds which show uniform mass segregation ($1.0 < \Lambda_3 < 1.5$), with standard deviation; certain clouds exhibit an increase in MSR at higher N . (*Bottom*) The MSR plots for the inversely mass segregated cloud, G0.316-0.201 ($\Lambda_3 < 0.75$), as well as the clouds which are uniformly distributed at $N = 3$ with $0.75 < \Lambda_3 < 1.0$.

within the clouds. The clouds which are shown in Figure 3 are truncated at $N = 20$ (unless otherwise stated) so that a crowding effect from the points is avoided.

The 2 clouds which display slightly stronger mass segregation, G0.489+0.010 and G1.602+0.018, are classified into the third case, see the top left panel of Figure 3. This case includes the Λ_3 values, without considering standard deviation, of ≥ 2.5 . We define the strongest mass segregation to occur when $\Lambda_3 \geq 2.5$ within standard deviation. The only region which exhibits an $\Lambda_3 \geq 2.5$ within standard deviation is G0.489+0.010 (Dust Ridge E&F).

5.1.1. Arithmetic Mass Segregation: Robust Catalog

For completeness, we also calculate the arithmetic MSRs for clouds containing at least 12 compact sources in the robust *CMZoom* catalog. We perform this calculation to better characterize the effect of removing low signal-to-noise source detections in our MSR calculations. As explained in Appendix A, we can not use $N = 3$ and $K = 500$ for clouds with only 12 compact sources, and we instead use $N = 4$ and $K = 450$. While there are variations in the MSR curves due to fewer compact sources within the clouds, the overall classifications of the analyzed clouds remain unaltered. These values are not reported.

5.2. Geometric Mass Segregation in the Clouds

The geometric MSR is designed to reduce the overall influence of the outlier compact sources while increasing sensitivity to milder levels of mass segregation, as compared to the arithmetic MSR (Olczak et al. 2011). The MSR curves that differ are seen to have outlier compact sources, such as G0.489+0.010 (see Figure 2 for its MST). Outlier compact sources are known to influence the specific shape of the arithmetic curve when included in the K random sets, due to increasing the average MST length. See Section 6.2 for more details.

Overall, we find that using the geometric MSR calculation does not change the mass segregation classifications for any of the *CMZoom* clouds. Shown in Figure 4 is the difference between the geometric and arithmetic methods for $N = 3$ and $N = 4$. We find a general agreement between the methods, and outlier compact sources (see Sections 4.1 and 6.2) are not so spatially separated as to widely affect the calculated MSR values.

We report the Λ_{MSR} and Γ_{MSR} for $N = 3$ and $N = 4$ for each cloud, along with the standard deviations, in Table 1.

5.3. Results of the Jeans Analysis

We follow the analysis of Sanhueza et al. (2019) by comparing the corrected core separations, ℓ_{corr} (not

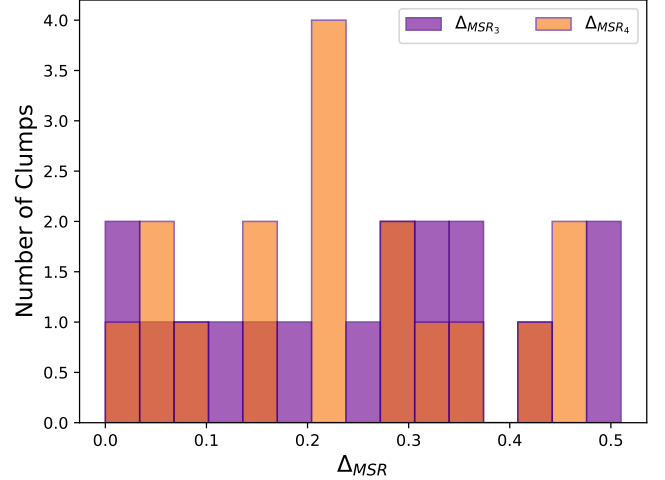


Figure 4. This histogram shows the difference (Δ_{MSR}) between the arithmetic and geometric MSRs for all 17 clouds that have at least sixteen compact sources, for $N = 3$ (purple) and $N = 4$ (orange).

$\bar{\ell}_{\text{corr}}$), of each region with λ_{th} to consider fragmentation at the cloud- and clump-scale. This comparison is performed with ratios of the core properties (ℓ_{corr} and M_{source}) to the calculated Jeans properties for the clumps and clouds. When the ratio is ≈ 1 , the core properties are similar to the predicted thermal or turbulent Jeans properties, indicating that the cores are consistent with either thermal or turbulent fragmentation. Ratios are performed for both the turbulent Jeans properties (length and mass) and the thermal Jeans properties (length and mass). These ratios can be found in Figure 5.

If we assume a dust temperature of 20 K, we find that the cloud-scale thermal ratios are ≥ 1 and the turbulent ratios are $\ll 1$ ⁴. This indicates that the source separations and masses are much smaller than what would be predicted under turbulent fragmentation. At the clump scale with the assumption of the $\sigma \propto R^{0.66}$ power law and the elliptical fit, we find that the turbulent ratios are still largely $\ll 1$, and that using the clump-scale properties in our analysis does not change our results. These calculations include the multiple H_2CO emission lines detected by Walker et al. (2024) in 6 of the clouds.

In summary, we find that, for most of the clouds analyzed, ℓ_{corr} is longer than the λ_{th} for the cloud, and not as long as the λ_{turb} for the cloud. Likewise, we find that M_{th} tends to be smaller than the compact source masses (M_{source}), and M_{turb} is generally larger than M_{source} . At the cloud scale, we observe that the source-separations

⁴ The median of the ratio $\frac{\bar{\ell}_{\text{corr}}}{\lambda_{\text{turb}}} \approx 1.07$, and of $\frac{M_{\text{turb}}}{M_{\text{source}}} \approx 2e4$.

Cloud Name	CN	Λ_3	Λ_4	Γ_3	Γ_4	$\bar{\ell}_{\text{corr}}$ (pc)	λ_{th} (pc)	$\lambda_{\text{turb}}^{\text{H}_2\text{CO}}$ (pc)	M_{th} ($M_{\odot}$)	$M_{\text{turb}}^{\text{H}_2\text{CO}}$ ($10^{+4} M_{\odot}$)
G1.670-0.130	16	1.85 ± 0.74	2.28 ± 0.71	$1.48 \pm_{0.62}^{3.58}$	$1.92 \pm_{0.53}^{4.39}$					
G1.651-0.050	12					1.72	$0.30 \pm_{0.23}^{1.01}$	$6.77 \pm_{1.98}^{2.80}$	$6.19 \pm_{6.03}^{234}$	$6.48 \pm_{1.89}^{2.68}$
G1.602+0.018	17	2.91 ± 1.06	2.50 ± 0.66	$2.93 \pm_{2.93}^{7.95}$	$2.07 \pm_{5.69}^{9.85}$	1.33	$0.32 \pm_{0.25}^{1.04}$	$2.12 \pm_{0.62}^{0.88}$	$6.83 \pm_{6.65}^{257}$	$2.92 \pm_{0.85}^{1.21}$
G1.085-0.027	14					1.34	$0.54 \pm_{0.41}^{1.76}$	$8.13 \pm_{5.75}^{11.5}$	$12.9 \pm_{12.6}^{488}$	$11.2 \pm_{7.92}^{15.8}$
G0.891-0.048	2					13.1	$0.50 \pm_{0.38}^{1.63}$	$21.4 \pm_{6.27}^{8.86}$	$12.7 \pm_{12.4}^{479}$	$99.1 \pm_{29.0}^{41.0}$
G0.891-0.048								$13.6 \pm_{3.99}^{5.64}$		$25.5 \pm_{7.48}^{10.5}$
G0.714-0.100	90	1.03 ± 0.43	1.38 ± 0.44	$1.16 \pm_{0.18}^{2.51}$	$1.44 \pm_{0.61}^{3.50}$	1.11	$0.38 \pm_{0.29}^{1.24}$	$20.2 \pm_{5.92}^{8.37}$	$10.2 \pm_{9.89}^{383}$	$151 \pm_{44.3}^{62.7}$
G0.714-0.100								$10.1 \pm_{2.96}^{4.18}$		$18.9 \pm_{5.54}^{7.84}$
G0.699-0.028	76	1.93 ± 0.91	2.44 ± 0.92	$1.61 \pm_{0.70}^{3.93}$	$1.93 \pm_{0.64}^{4.50}$	0.92	$0.20 \pm_{0.16}^{0.66}$	$4.62 \pm_{3.27}^{6.54}$	$5.43 \pm_{5.29}^{204}$	$6.37 \pm_{4.51}^{9.01}$
G0.699-0.028								$6.94 \pm_{4.91}^{9.82}$		$21.5 \pm_{15.2}^{30.4}$
G0.619+0.012	23	0.84 ± 0.30	0.93 ± 0.27	$0.72 \pm_{0.42}^{1.86}$	$0.76 \pm_{1.66}^{3.18}$					
G0.489+0.010	32	4.47 ± 1.59	3.96 ± 1.15	$3.50 \pm_{1.53}^{8.53}$	$3.28 \pm_{2.26}^{8.83}$	0.94	$0.30 \pm_{0.22}^{0.96}$	$11.2 \pm_{7.90}^{15.8}$	$7.87 \pm_{7.66}^{297}$	$42.7 \pm_{30.2}^{60.4}$
G0.412+0.052	18	1.20 ± 0.40	1.48 ± 0.31	$0.96 \pm_{0.76}^{2.68}$	$1.23 \pm_{1.60}^{4.07}$	1.05	$0.32 \pm_{0.25}^{1.05}$	$11.9 \pm_{8.43}^{16.8}$	$9.02 \pm_{8.79}^{340}$	$45.6 \pm_{32.2}^{64.4}$
G0.380+0.050	18	1.07 ± 0.40	0.92 ± 0.23	$1.06 \pm_{1.00}^{1.12}$	$0.84 \pm_{0.35}^{2.00}$	0.74	$0.26 \pm_{0.20}^{0.87}$	$8.82 \pm_{6.24}^{12.5}$	$7.42 \pm_{7.23}^{280}$	$27.3 \pm_{19.3}^{38.6}$
G0.380+0.050								$2.94 \pm_{2.08}^{4.16}$		$1.01 \pm_{0.72}^{1.43}$
G0.340+0.055	16	1.28 ± 0.44	1.39 ± 0.42	$1.35 \pm_{0.10}^{2.81}$	$1.49 \pm_{1.24}^{4.22}$	1.07	$0.43 \pm_{0.33}^{1.39}$	$15.0 \pm_{10.6}^{21.3}$	$13.1 \pm_{12.7}^{494}$	$57.6 \pm_{40.7}^{81.5}$
G0.326-0.085	21	0.90 ± 0.39	0.88 ± 0.33	$0.60 \pm_{2.67}^{3.87}$	$0.57 \pm_{1.50}^{2.64}$	1.12	$0.72 \pm_{0.16}^{3.08}$	$26.9 \pm_{19.1}^{38.1}$	$24.1 \pm_{0.62}^{935}$	$125 \pm_{88.2}^{177}$
G0.316-0.201	17	0.69 ± 0.23	0.75 ± 0.18	$0.58 \pm_{0.30}^{2.04}$	$0.75 \pm_{0.30}^{1.81}$	0.76	$0.63 \pm_{0.48}^{2.06}$		$18.5 \pm_{18.0}^{700}$	
G0.253+0.016	16	1.11 ± 0.51	1.38 ± 0.51	$0.90 \pm_{0.55}^{1.46}$	$1.25 \pm_{0.79}^{1.98}$	1.02	$0.69 \pm_{0.53}^{2.25}$	$18.6 \pm_{13.2}^{26.4}$	$22.9 \pm_{22.3}^{866}$	$45.7 \pm_{32.2}^{64.6}$
G0.253+0.016								$34.9 \pm_{24.7}^{49.4}$		$301 \pm_{213}^{425}$
G0.145-0.086	9					1.37	$0.31 \pm_{0.24}^{1.03}$	$11.3 \pm_{2.99}^{4.23}$	$9.21 \pm_{8.98}^{348}$	$31.6 \pm_{9.27}^{13.1}$
G0.106-0.082	15					0.58	$0.31 \pm_{0.24}^{1.02}$	$10.2 \pm_{3.32}^{4.70}$	$9.21 \pm_{8.98}^{348}$	$43.4 \pm_{12.7}^{17.9}$
G0.070-0.035	9					2.63	$0.57 \pm_{0.44}^{1.87}$	$21.4 \pm_{15.1}^{30.3}$	$19.1 \pm_{18.6}^{723}$	$99.1 \pm_{700}^{140}$
G0.068-0.075	22	1.11 ± 0.39	0.77 ± 0.22	$1.25 \pm_{0.25}^{6.19}$	$0.80 \pm_{0.36}^{1.82}$	0.80	$0.25 \pm_{0.19}^{0.81}$	$9.12 \pm_{2.67}^{3.78}$	$6.91 \pm_{6.73}^{261}$	$34.9 \pm_{10.2}^{14.4}$
G0.001-0.058	54	1.13 ± 0.49	1.13 ± 0.40	$0.92 \pm_{0.82}^{1.04}$	$0.88 \pm_{0.62}^{1.26}$	0.83	$0.30 \pm_{0.23}^{0.96}$	$13.3 \pm_{3.89}^{5.49}$	$9.44 \pm_{9.20}^{356}$	$85.7 \pm_{25.1}^{35.5}$
G359.948-0.052	69	0.90 ± 0.36	1.10 ± 0.39	$0.76 \pm_{0.01}^{1.55}$	$0.98 \pm_{1.37}^{3.35}$					
G359.889-0.093	48	1.06 ± 0.43	1.35 ± 0.45	$1.10 \pm_{0.75}^{1.62}$	$1.19 \pm_{0.54}^{2.60}$	1.01	$0.33 \pm_{0.25}^{1.06}$	$14.5 \pm_{4.24}^{5.99}$	$9.13 \pm_{8.89}^{345}$	$79.7 \pm_{23.3}^{33.0}$
G359.648-0.133	17	1.84 ± 1.18	2.15 ± 1.08	$1.14 \pm_{1.38}^{3.68}$	$1.38 \pm_{1.67}^{3.35}$	1.43	$0.83 \pm_{0.64}^{2.72}$	$13.6 \pm_{3.97}^{5.61}$	$26.7 \pm_{26.0}^{1008}$	$12.9 \pm_{3.79}^{5.37}$
G359.615-0.243	7					0.90	$0.59 \pm_{0.45}^{1.92}$	$4.15 \pm_{1.21}^{1.72}$	$18.0 \pm_{17.5}^{680}$	$0.63 \pm_{0.18}^{0.26}$
G359.484-0.132	6					2.71	$0.78 \pm_{0.60}^{2.56}$	$23.1 \pm_{6.76}^{9.56}$	$25.0 \pm_{24.4}^{947}$	$56.5 \pm_{16.5}^{23.4}$

Table 1. This table shows the arithmetic (Λ_{MSR}) and geometric (Γ_{MSR}) MSRs with corresponding uncertainties calculated at $N_{MST} = 3$ and 4 for each eligible *CMZoom* cloud (with source number (CN) ≥ 16), as well as the calculated $\bar{\ell}_{\text{corr}}$ and cloud-scale Jeans lengths and masses for each cloud included in both [Battersby et al. \(2025b\)](#) and [Walker et al. \(2024\)](#); the clouds which are not included are G1.670-0.130, G0.619+0.012, and G359.948-0.052. Certain clouds have multiple row entries in this table since they have multiple linewidths reported in [Walker et al. \(2024\)](#) γ_{MSR} is not explicitly reported, but instead the 1σ deviation of Γ_{MSR} and γ_{MSR} . MSRs are calculated using compact sources from our modified version of the complete *CMZoom* catalog. All MSRs have been rounded to 2 decimal places, as well as the Jeans lengths and masses. The turbulent Jeans masses are reported in units of $10^{+4} M_{\odot}$.

(ℓ_{corr}) are more consistent with thermal fragmentation than turbulent fragmentation, when in the presence of high turbulence present in the CMZ.

These reported results do not include the clump-scale analysis (see Figure 5 for the smaller-scale results). However, as seen in Figure 5, both the cloud-scale and the clump-scale thermal analyses tend to favor fragmentation and gravitational contraction, while the turbulent analyses tend not to.

6. DISCUSSION

As shown in Figure 3, the majority of the regions analyzed in this paper exhibit either inverse mass segregation or a nearly uniform mass distribution, with only 5 classified as mass segregated. This indicates that mass segregation is not necessarily common in these regions of the CMZ at this resolution and size scale.

6.1. Mass Segregation and Star Formation

Cloud Name	CN	MS	MS _{BGS}	SF _{Rob}	SF _{Amb}	SF _{3DCMZII}
G1.670-0.130	16	Yes	Yes	Yes	Yes	—
G1.602+0.018	17	Yes	Yes	Yes	Yes	Yes
G0.699-0.028	76	Yes	Yes	—	—	Yes* [◊]
G0.489+0.010	33	Yes [†]	Yes [†]	Yes*	Yes*	Yes*
G359.648-0.133	17	Yes	Yes	No	No	—
G0.380+0.050	18	Un.	Yes	Yes*	Yes	Yes* [‡]
G359.889-0.093	48	Un.	Yes	Yes*	Yes*	Yes [‡]
G0.326-0.085	21	Un.	Yes	No	Yes	—
G0.619+0.012	23	Un.	Un.	Yes*	Yes*	—
G0.001-0.058	55	Un.	Un.	Yes	Yes	Yes
G0.412+0.052	18	Un.	Un.	No	Yes	Yes
G0.714-0.100	90	Un.	Un.	No	Yes*	Yes*
G0.340+0.055	16	Un.	Un.	No	No	Yes [◊]
G359.948-0.052	69	Un.	Un.	—	—	—
G0.068-0.075	23	Un.	No	Yes	Yes	Yes
G0.253+0.016	17	Un.	No	No	Yes	Yes [◊]
G0.316-0.201	17	No	Un.	No	No	—

Table 2. In this table, we compare mass segregation for the 17 applicable *CMZoom* regions (with CN ≥ 16) with the presence of star-formation tracers as derived and shown in Tables 2 and 3 of Hatchfield et al. (2024) (the columns labeled “SF_{Rob}” and “SF_{Amb}” here) as well as in Table 4 of Battersby et al. (2025b) (“SF_{3DCMZII}”), with “SF” indicating that the region is known to be star-forming. [†] indicates clouds with MSR > 4 . * indicates star-formation rates (SFR) $\geq 1.0\text{e-}02 M_{\odot}\text{yr}^{-1}$. [‡] indicates clouds where the SFR is calculated using a uniform T of 50 K rather than the Herschel Hi-GAL temperature (≈ 20 K). [◊] indicates SFR estimates calculated using free-fall time. Distinct MSRs > 1.5 are represented by “Yes”, MSRs between 0.75 and 1.5 with “Un.” (“Uniform”), and MSRs < 0.75 with “No”. MS_{BGS} is the MSR as derived from the compact source masses with the background flux subtracted, see Section 6.2. G0.699-0.028 and G359.948-0.052 are excluded from (Hatchfield et al. 2024) due to uncertainties in the mass estimates, and G359.948-0.052 is excluded from Battersby et al. (2025b) due to a lack of background-subtracted structure.

As discussed briefly in the Section 1, the CMZ might be underproducing stars given the amount of dense gas it has overall. Battersby et al. (2025b) finds that the calculated SFRs, while low for the entire CMZ, are expected for the gas contained in the clouds which are currently forming stars, and concludes that much of the gas and dust in the CMZ is simply not engaged in star-formation. We use the SFRs calculated in Hatchfield et al. (2024) to determine the relationship between the mass segregation of our clouds and their star formation activity, as Hatchfield et al. (2024) covers a complete sample of clouds throughout the CMZ.

We find that four clouds consistently show agreement between observed mass segregation and detected star-formation with the non-background subtracted compact sources, and that one cloud shows inverse mass segregation and no star-formation tracers. With the background subtracted compact sources, seven clouds show both mass segregation and star formation, while two clouds show neither.

There is no obvious correlation between the mass segregation observed in this work and the evolutionary stage of the clouds analyzed. G0.489+0.010 exhibits

the highest degree of mass segregation in this work, while G0.380+0.050 exhibits a roughly uniform level of mass segregation, and G0.412+0.052 (Dust Ridge D) a slightly higher MSR of ≈ 1.20 ; the first two clouds are considered to be evolutionarily young due to the presence of star formation tracers (Lu et al. 2019) (Barnes et al. 2019; Walker et al. 2017) and the latter-most even younger (Barnes et al. 2019). G0.699-0.028, which presents a lower and more complex MSR than G0.489+0.010, is likely more evolved due to a large population of protostellar cores (Ginsburg et al. 2018). The mostly quiescent G0.253+0.016 cloud, which displays uniform mass segregation, is considered to be at an early evolutionary stage (Walker et al. 2021; Mills et al. 2015; Immer et al. 2012), with a steeper CMF than that of the actively star-forming G0.699-0.028 (Kinman et al. 2024). G1.602+0.018 displays mass segregation but has not been extensively studied as to its current evolutionary state. The other clouds in this work are too ambiguous with respect to their mass segregation values, and provide little to no statistical significance to a correlation between mass segregation and star formation (or mass segregation and evolutionary stage).

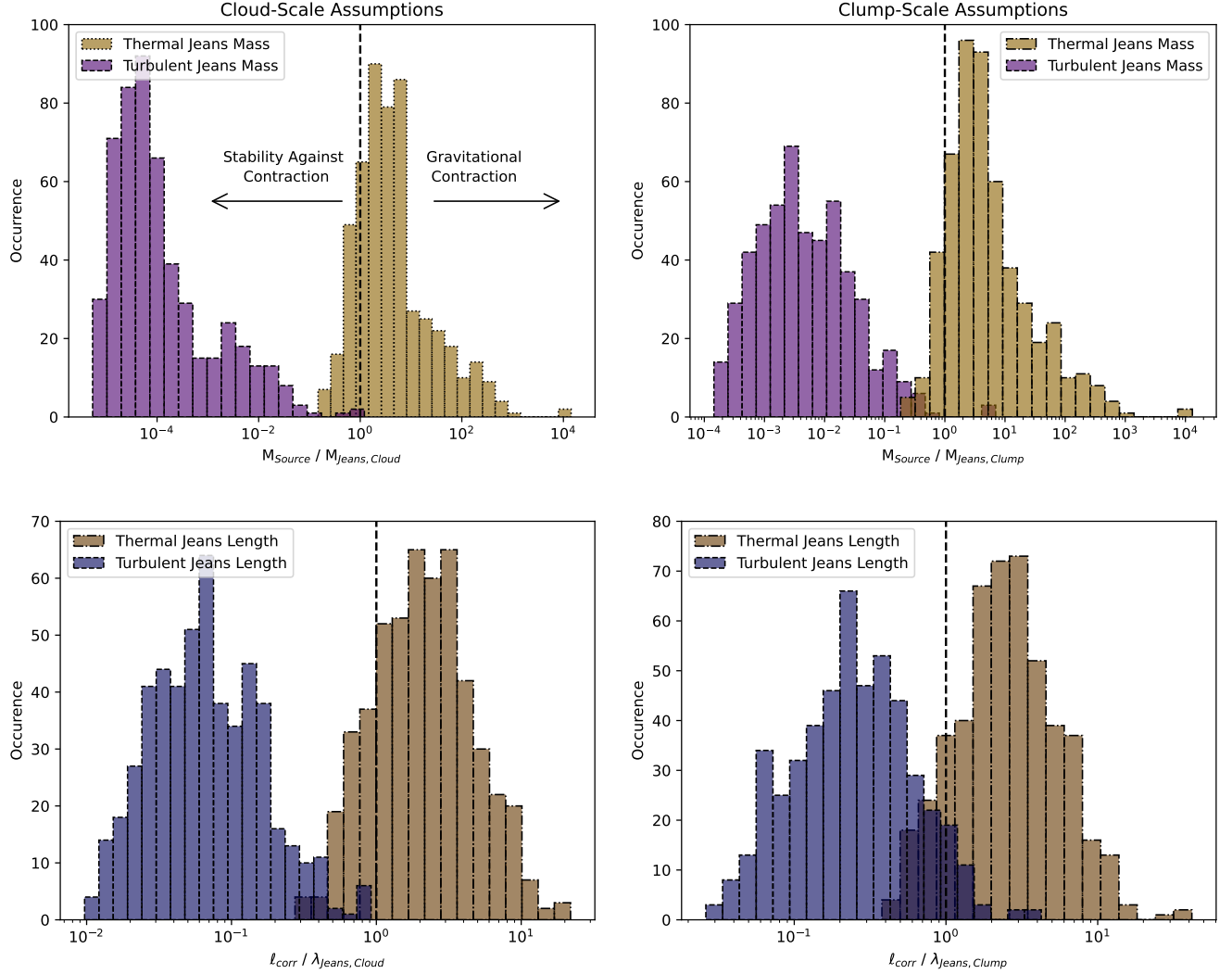


Figure 5. This figure compares compact source separations and masses to the cloud-scale Jeans calculations (lengths and masses) and clump-scale Jeans calculations (lengths and masses). We conclude that the thermal Jeans calculations more closely match the measured values, and also show the potential for gravitational contraction and fragmentation of the sources at small scales while turbulence dominates at large scales. The data is presented as a ratio of values, such that a ratio of 1:1 represents the equilibrium between the gravitational force and the thermal or turbulent pressures of the gas. The x-axes are scaled by $\log_{10}$ for concision. (*Top Left*) The histogram of the compact source masses from each cloud as a ratio with the thermal and turbulent Jeans masses for the cloud to which the respective compact sources belong. The dashed line marks equilibrium, a ratio of 1:1. (*Bottom Left*) The compact source separations (deprojected minimum spanning tree edge lengths, ℓ_{corr}) in a ratio with the thermal and turbulent Jeans lengths for the cloud to which the respective compact sources belong. (*Top Right*) The compact source masses in a ratio with the thermal and turbulent Jeans mass for the respective clump. (*Bottom Right*) The compact source separations in a ratio with the thermal and turbulent Jeans length for the respective clump.

In summary, we do not see clear evidence of a correlation between mass segregation and SFR in this work. We find that mass segregation is not the best indicator for current star-formation within the clouds, and that clouds which are actively star-forming are not necessarily more likely to be primordially segregated.

6.2. Possible Sources of Error

There are multiple factors that can contribute to uncertainty in our analysis that come from both the data and our methods.

1. **Sources in 2D** One potential source of error is our definition of a “compact source”. These are defined based on their relative column density compared with the background continuum. However, these clouds are observed as two-dimensional on the sky, and are three-dimensional in reality. This can allow multiple compact sources to “stack” on top of each other in the line-of-view. Further, the masses of each compact source are derived from the overall flux. Multiple compact sources that are stacked on top of each other will appear to have a higher flux than if they had been spread out individually. This has the potential to affect not only the counted number of compact sources in each cloud but also the masses of those compact sources. This might explain the unusually high masses of the compact sources in the G0.699-0.028 cloud. Further, if the compact sources are possibly under-counted due to the method of observation used, the MSTs drawn from the observed compact sources do not properly represent the morphology of the cloud, and this in turn affects the derived MSRs. Differences in dust temperature also have the potential to affect the calculated MSRs (as briefly discussed in Section 4.1 and reported in Hatchfield et al. 2020).

2. **Imaging Artifacts** Furthermore, the noise present in the SMA data can result in possible false positive detections, particularly along the defined edges of the regions. Because of this, as previously stated in Section 2, we use a version of the complete catalog with the edge sources removed. We also acknowledge that sources might exist beyond the boundaries of our defined regions, however, the *CMZoom* observations were designed to cover the densest clouds in the CMZ where star formation would be located, so we consider it unlikely that any significant sources are outside of the regions investigated here.

The G0.699-0.028 and G359.948-0.052 regions suffer from imaging artifacts and can be expected to have higher local rms noise levels as a result. Since the *CMZoom* catalog accounts for the local noise at the location of each leaf, we would expect these regions to be pruned more stringently for inclusion in either the robust or complete catalogs than in other regions that may lack these imaging artifacts.

3. **Outlier Sources** We find that very few regions have statistical outliers, where there are MST edge lengths that vary significantly from the mean edge length, while some regions contain none. Of the sources which may be outliers, as often as not, they are centrally located within the region, or otherwise located such that their removal from the set results in the creation of more severe outliers. For our analysis, we are mainly concerned with potential false detections from noise, which may produce “massive” outliers that are not centrally located. This is the case with regions G1.602+0.018 and G359.648-0.133 in the complete catalog, where the massive outliers lie at the edge of the map. The subsequent removal of these edge sources in the reduced version of the complete catalog allows these regions to become more mass segregated.

4. **Complete vs. Robust Catalog** As mentioned in Section 2.2, we primarily use the complete *CMZoom* catalog in this work (although we do briefly discuss the differences in results with the *CMZoom* robust catalog, in Section 5.1.1). Our choice of catalog can potentially impact our resulting MST analysis. The complete catalog likely contains “false positive” detections from the noise in the observations, whereas the robust catalog may be removing real, low-brightness sources. These differences propagate out to the results from our analysis, and the truth of the clouds can be assumed to lie somewhere in-between. Hatchfield et al. (2020) consider the completeness of both catalogs, and find that the complete catalog is 95% complete at 50 $M_{\odot}$, while the robust catalog is 95% complete at 80 $M_{\odot}$. Furthermore, as discussed briefly in Section 2, there is unlikely to be very many false positives among the lower mass cores of the complete catalog, with the robust catalog already containing most of the mass within the regions. Hatchfield et al. (2020) compared the SMA data for a few key regions with corresponding ALMA data in order to better understand the veracity of the catalog sources, with respect to the fila-

mentary structures. The authors find that there is good agreement between the images, while the G0.699-0.028 SMA image contains some sources not found in the ALMA image. This discrepancy could be due to the fact that the G0.699-0.028 region contains both the highest noise and the highest flux found within the *CMZoom* catalog. Missing detections or including false detections from the catalog will impact our results, however the *CMZoom* catalog included high flux thresholds in the creation of the catalog and considered the influence of edge sources in order to mitigate the impact of false detections. Most missing detections will be lower mass sources that would have less impact on the MSR calculation. Additionally, we show that the MSR classifications for each region changes minimally when using the robust catalog, as shown in Section 5.1.1.

5. **Measurement Errors** When we use the background subtracted mass for each source, the position of the most massive compact source in a cloud may change. This affects the MSR directly. We observe that some clouds, such as G359.948-0.052 near Sgr A*, become more mass segregated with the background subtracted masses. The majority of the clouds analyzed do not exhibit any significant difference in the MSR values when the background subtracted masses are used compared to the MSR values when the background is not subtracted from the mass estimations. The clouds for which there is a significant difference are shown in Figure 7. Based on this evaluation, we do not consider the use of the background subtracted masses to change our conclusions regarding our MST-based analyses.

Additional sources of error come from the analysis with the thermal and turbulent Jeans lengths and masses. One source of error is the median dust temperature used in the calculation of the thermal Jeans lengths and masses, and stated in Section 3 to be roughly 10%. This error is further complicated by systematic effects including the variations of temperatures along the line of sight, within a beam, or variations between the small scales of the SMA and the larger scales of Herschel (see: Battersby et al. 2020; Battersby et al. 2011). Higher resolution studies of the CMZ, such as Mills & Morris (2013) and Ginsburg et al. (2016), find higher temperatures associated with smaller-scale structures located within the clouds.

Another source of error belongs to the densities from the Herschel Hi-GAL table, which are considered to have uncertainties of a factor of 2 (also discussed in Section 3). The last source of error is from the compact source masses themselves, as discussed in Battersby et al. (2020), which are considered also to have uncertainties of roughly a factor of 2 (see Battersby et al. 2020 for more detail).

6. **Reliability of the *CMZoom* catalog** The compact sources are treated as point sources in this work, which raises concerns over the validity of calculations involving their separations, particularly given that no two sources can physically be closer than their radii. The dendrogram leaves have hard boundaries which do not overlap, resulting in a limit to how close the leaves can be to each other. This could potentially result in false negatives, as in a hypothetical where projection causes two otherwise distinct sources to be treated as one. Hypothetical sources which are close enough to each other to raise concerns over projection and may be considered one massive object by the dendrogram algorithm would be likely to contribute toward shorter MST edge lengths. We do not consider this possible error to significantly affect our MSR analysis, since a hypothetical massive source which causes a region to be considered mass segregated, if instead considered to be multiple smaller sources, would nonetheless contain a larger fraction of mass than the other sources in the region, and therefore, the mass is still segregated in that part of the region.

7. CONCLUSIONS

The Central Molecular Zone has long been considered to have unusually low levels of current star-formation given its extreme physical conditions. The *CMZoom* survey has allowed us to investigate mass segregation across the densest star forming clouds in the CMZ in a consistent and comparable manner, in order to better understand the differences and similarities between the clouds, rather than focusing on only one cloud at a time.

We use MST-based analyses to quantify the spatial distribution and mass segregation of compact sources identified in the *CMZoom* survey. We also investigated how mass segregation correlates with the presence of star formation tracers and thus the evolutionary stage of a cloud. We summarize our findings below:

1. We compare the MSRs of the clouds to each other to better understand the CMZ as a whole, rather than cloud by cloud as it has been considered pre-

viously. We find that one-third of the clouds surveyed exhibit mass segregation, while the remaining two-thirds exhibit either inverse mass segregation or uniformity.

2. We calculate both the arithmetic and geometric mass segregation ratios of the clouds included in the *CMZoom* survey with more than 15 compact sources. We find that the geometric MSRs generally present the same curves as the arithmetic MSRs, as expected, and in clouds with outliers, the effect on the MSR is small.
3. We find that, despite the widespread presence of turbulence in the CMZ, the clouds included in this work appear to be fragmenting according to the thermal Jeans length and mass, at the observed source resolution of ≈ 0.1 pc. This result is consistent with those found in other CMZ studies that observed fragmentation at ≤ 0.1 pc scales (Lu et al. 2020; Kruijssen et al. 2019; Pokhrel et al. 2018).
4. We briefly consider a possible link between the MSRs calculated in this work and the star-formation rates presented in Hatchfield et al. (2024), as well as the presence of star formation tracers as an evolutionary indicator for the clouds. We do not find a strong correlation between the MSRs and the presence of star formation.

This is the CMZ-wide study of the spatial distribution and mass-segregation of compact sources, where previous studies have considered individual clouds. Ul-

timately, we find that there is no strong correlation between the mass segregation for star-forming regions in the CMZ and their evolutionary state at 0.1 pc scales; however, higher resolution studies towards a large and representative sample of regions will be needed to see if this is still true at core (1000 au) scales.

8. ACKNOWLEDGMENTS

We acknowledge the use of the *CMZoom* Survey, first published in 2020, without which this work could not exist. S. Schuler, J. Wallace, and C. Battersby gratefully acknowledge funding from the National Science Foundation under Award No. 2108938. S. Schuler and C. Battersby were also supported by the National Science Foundation Award No. 2206510 and J. Wallace and C. Battersby were supported on Award No. CAREER 2145689. C. Battersby also gratefully acknowledges support from the National Aeronautics and Space Administration through the Astrophysics Data Analysis Program under Award “3-D MC: Mapping Circumnuclear Molecular Clouds from X-ray to Radio,” Grant No. 80NSSC22K1125. S.Z. acknowledges support from the Strategic Priority Research Program of the Chinese Academy of Sciences (CAS) Grant No. XDB0800303. RAG acknowledges funding support from: NASA ADAP awards NNX11AD14G, NNX13AF08G, NNX15AF05G, and NNX17AF24G; NSF AAG grants 1636621, 1812747, and 2107705; NASA-USRA SOFIA grants 05-0181, 07-0225, and 08-0181; NASA-JPL/Caltech Spitzer grants 1373081, 1424329, and 1440160, and Herschel grant 1489384.

APPENDIX

A. DISCUSSION OF K VALUES, AND THE ARITHMETIC MSR

As stated in Section 4.1, we choose a K value of 500. This choice comes from the behavior of the Arithmetic MSR when calculated on randomly uniformly distributed clouds. Allison et al. (2009) state that there is a roughly Gaussian dispersion associated with the mean lengths of the random MSTs (quantifiable by the standard deviation $\langle \ell \rangle \pm \sigma$). This is due to the specific random sets of compact sources that are chosen and averaged to make the numerator of the ratio. With an increase of K, especially if the sets are randomly chosen without replacement, a greater fraction is taken of the total number of random sets available. Each time that the MSR calculation is run with replacement sampling, it is possible for the same random sets to be sampled again, which can produce a less accurate calculation of the MSR.

When sampling the random sets without replacement, it is important to only sample up to the number of available unique combinations. Therefore, setting a specific K will dictate the smallest N_{MST} and lowest compact source number (CN) that can be used. We find that a K value of 500 is adequate for our analysis in this work, as it allows us to have a CN as low as 16 and an N_{MST} as small as 3. See the left hand panel of Figure 6 for the comparison between K values for the same randomly generated cloud with 20 compact sources and $N_{MST} = 4$.

Allison et al. (2009) recommend a K of 500 or 1000 for low N_{MST} and a K of at least 50 for high N_{MST} . This is due to the fact that as N_{MST} approaches the total CN of a given uniformly distributed cloud, the random MST lengths do not differ from each other greatly, allowing for there to be smaller deviations between them. This can be seen in Table 3, where the standard deviation of a cloud with 25 compact sources, $N_{MST} = 4$, and $K = 500$ is similar to the same cloud with $N_{MST} = 13$ and $K = 50$. The same holds for other clouds of different sizes. We choose $K = 500$ for our clouds regardless of N_{MST} to obtain results of the same or better precision.

The established definition of inverse mass segregation describes when a cloud exhibits MSR values lower than 1.00, and is observed only to fit HMSF 0. By our definition, these values between 0.75 and 1.00 fall within the 1σ deviation of a uniformly distributed cloud (see Appendix A), and are neither truly inversely mass segregated nor mass segregated. This can be seen in the right panel of Figure 6, where the plotted MSR distribution for 1000 unique, randomly uniformly distributed clouds at a K of 500 shows that, for randomly assigned mass values within the simulated regions, the MSR follows a log-normal distribution with the majority of the MSR values falling between 0.75 and 1.50. We therefore take these values as bounds for MSRs indicative of uniformly distributed regions.

A.1. Confidence of the Arithmetic MSR

The mass-segregated clouds, or the MSR values themselves, are fairly confident per region in the *CMZoom Survey*. This is tested by taking a region at a certain N and calculating the MSR for that region at that N 1000 separate times. These 1000 MSR values are then looked at to see how many fall into a different classification than where the region was originally placed. All regions at $N = 3$ and $N = 4$ consistently have MSR values within the same classification. At higher N values, the morphology of the particular region plays a part in what fraction of the 1000 iterations will fall into a different class at that N , however, the MSR naturally moves through classes as it approaches 1. This suggests that any effect on the MSR by possible outliers is negligible.

Maschberger & Clarke (2011) suggests the use of median edge lengths rather than mean edge lengths in the calculation of the MSR in order to minimize the influence of outliers. In our tests of Maschberger & Clarke (2011)’s method compared to Allison et al. (2009)’s method, we find that at $N = 3$ for the reduced catalog, the “median” MSR, when iterated 1000 times, tends to have greater standard deviation, as can be expected by the use of the median. Additionally, for G359.648-0.133, Maschberger’s method results in 11 out of 1000 iterations (1.1%) falling into a different category.

We test the sensitivity of the arithmetic MSR to “leave-one-out”, where each compact source is excluded from the sample. We consider the use of the complete catalog for two regions: G1.602+0.018 and G0.489+0.010, due to the complete catalog’s higher population of cores and potential outliers. The former region contains a “massive” compact source close to the edge of the region, which is included in the minimum spanning tree for the most massive sources. When this source is removed, the MSR changes, otherwise, the MSR is consistently uniform with the removal of the other sources. The latter region contains two sources near to its edges, however, these sources are not massive, and the exclusion of each source does not affect the MSR. When these two sources are excluded one at a time, the MSR changes to reflect the average shortening of the random minimum spanning trees; this change is not significant overall due to the high value of the MSR to begin with.

B. MSRS FOR BACKGROUND-SUBTRACTED SOURCES

As previously stated, we use the non-background-subtracted sources for our calculations in this paper. We calculate the MSRs for the clouds with the background-subtracted sources and observe very few differences between the results. Only four clouds, seen in Figure 7, have a distinct difference between the respective arithmetic calculations.

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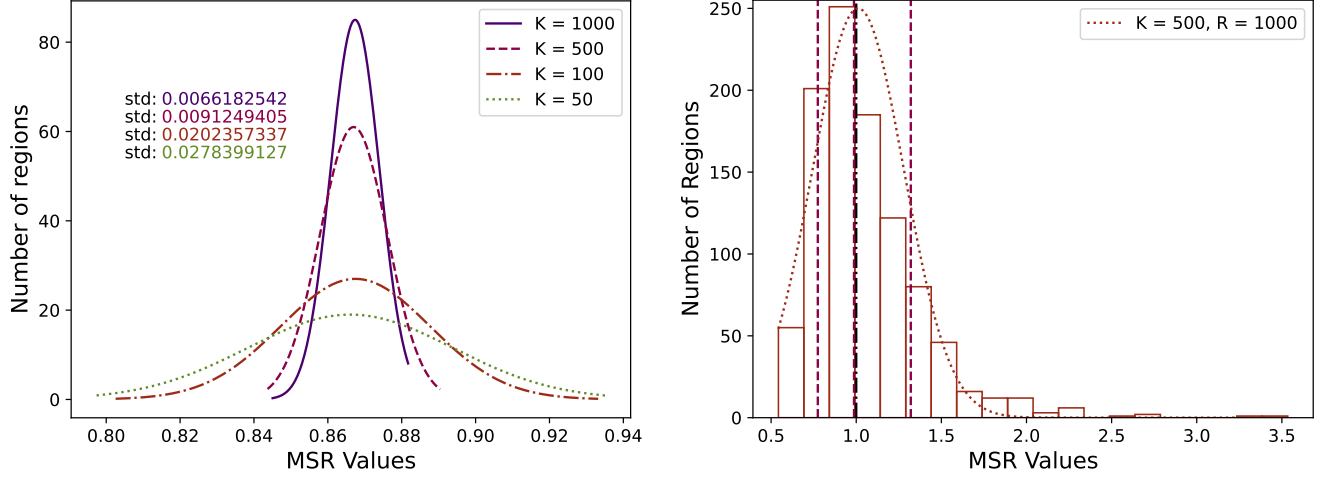


Figure 6. (*Left*) The distribution of MSR values for an ensemble of randomly uniformly distributed simulated clouds showing a decrease in the standard deviation with an increase in K , with the standard deviation listed on the left-hand side of the panel. The same cloud was iterated over 200 times for each chosen K with $N_{MST} = 4$ and a total compact source number of 20. The dashed lines are Gaussian approximations of the spread of the MSR values for each K . The figure is centered at an MSR of 0.87 because that value is where the majority of the calculated MSRs naturally fell — 0.87 is the median MSR for this particular randomly uniformly distributed cloud. (*Right*) The log-normal nature of the MSR distribution for 1000 randomly uniformly distributed clouds, at $K = 500$. The dashed vertical lines mark the 1σ percentages, and the dashed-dotted black line marks $MSR = 1.0$.

CN	N_{MST}	K Value	σ
25	4	50	0.0278
25	4	100	0.0202
25	4	500	0.0091
25	4	1000	0.0066
25	13	50	0.0099
25	13	100	0.0065
25	13	500	0.0029
25	13	1000	0.0022
200	32	50	0.0080
200	32	100	0.0065
200	32	500	0.0026
200	32	1000	0.0018
200	100	50	0.0037
200	100	100	0.0025
200	100	500	0.0011
200	100	1000	0.0008

Table 3. In this table are reported the standard deviations for unique randomly uniformly distributed simulated clouds of a certain CN and N , with varying K s. The unique clouds have been sampled 200 separate times in order to determine the standard deviations.

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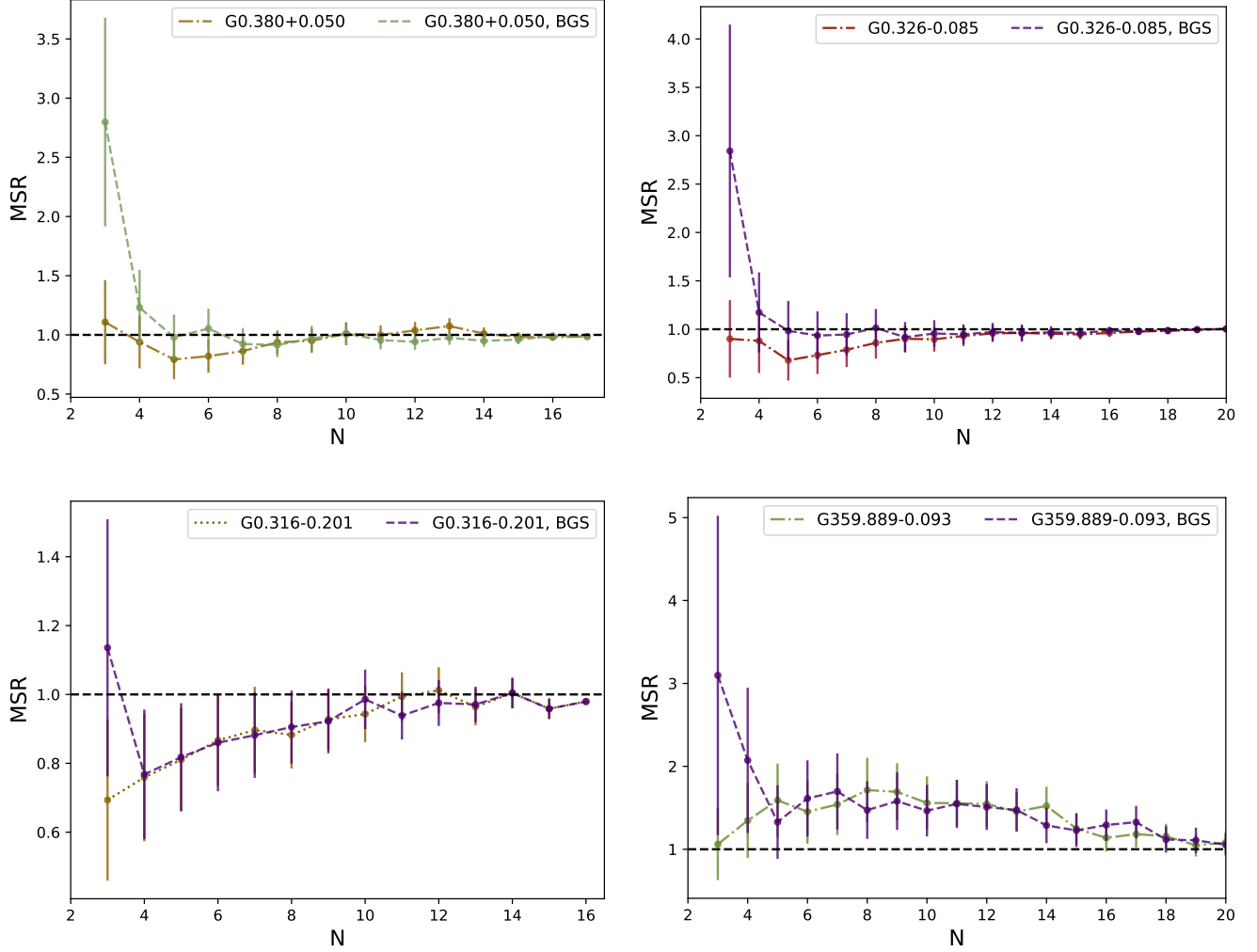


Figure 7. The arithmetic MSR plots for 4 clouds which display the greatest difference between the background-subtracted sources (BGS) and the non-background-subtracted sources. N is limited to $N = 20$ due to the tendency of the MSR to “level” at $\text{MSR} = 1$ for $\frac{N_{\text{total}}}{N_{\text{MSR}}} \rightarrow 1$. The dotted black line indicates $\text{MSR} = 1$.

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